Practice Problem Set 1

ECON 308 – International Economic Relations

Fall 2025

1. Explain why country size (GDP) and geographic distance are strong predictors of bilateral trade flows. Provide at least one microeconomic channel for each.
2. Define "home bias" in trade. Describe one empirical method for detecting a border effect in a gravity regression and one interpretation of finding a large border effect.
3. Why do small economies often exhibit higher trade-to-GDP ratios than large economies? Include at least one scale or variety argument.
4. Chapter 2 highlights both inter-industry and intra-industry trade. Briefly differentiate them and give one example of each between advanced economies.
5. Define both. Explain, using unit labor requirements, how a country can gain from trade even if it has an absolute disadvantage in *both* goods.
6. In the Ricardian model with free trade, what pins down relative wages across countries? Explain the role of world prices and specialization.
7. Distinguish between the production possibility frontier (PPF) and the consumption possibility frontier (CPF) under autarky versus free trade. Why does the CPF typically lie outside the PPF with trade?
8. Briefly explain why, under perfect competition, autarky relative prices equal unit labor cost ratios, and how opening to trade changes the mapping from technology to prices and wages.

9. The (log) gravity model is

$$\ln T_{ij} = \alpha + \beta_1 \ln Y_i + \beta_2 \ln Y_j - \gamma \ln D_{ij},$$

with T_{ij} bilateral exports from i to j, Y GDP, and D great-circle distance (km). Suppose $\beta_1 = \beta_2 = 1$, $\gamma = 1$, and $e^{\alpha} = 10^{-6}$ so that

$$T_{ij} = 10^{-6}\,\frac{Y_i Y_j}{D_{ij}}.$$

Consider the following economies (values in billions of US dollars; distances in km):

Country	Y (\$ bn)
A	2,000
B	1,000
C	500

	A	B	C
A	–	6,000	1,500
B	6,000	–	2,000
C	1,500	2,000	–

(a) Compute the predicted bilateral exports T_{AB}, T_{AC}, and T_{BC}.

(b) By what *percent* does T_{AB} change if D_{AB} falls by 10%? (Use the elasticity implied by the model.)

(c) If both Y_A and Y_B rise by 5%, what is the approximate percentage change in T_{AB}?

(d) Suppose the true T_{AB} observed in data is \$1.30 billion. Compute the (log) residual $\ln T_{AB}^{\text{obs}} - \ln T_{AB}^{\text{pred}}$ and provide one economic interpretation of a positive residual.

10. You are given the following national accounts (all in billions of domestic currency) and nominal GDP:

Country	Exports	Imports	GDP	Population (mn)
D	120	100	1,000	50
E	400	350	2,500	200
F	70	110	600	10

(a) Compute each country's openness ratio $(X + M)/\text{GDP}$ (in %).

(b) Compute *per capita* exports and imports for each country.

(c) Rank countries by openness and briefly explain why small economies often have higher openness ratios than large ones.

11. Extend the gravity model in Q1 by multiplying predicted trade by $(1+\lambda_{CL})$ if a common language is shared and by $(1 - \lambda_{BOR})$ if there is an international border friction.[1] Let $\lambda_{CL} = 0.20$ and $\lambda_{BOR} = 0.30$. Countries A and B share a common language; A and C do not; all pairs are separated by an international border.

(a) Starting from your baseline T_{ij} in Q1(a), compute adjusted T_{AB}, T_{AC}, and T_{BC}.

(b) By what percent does the common language increase T_{AB} relative to the case with no language effect?

(c) Holding other parameters fixed, solve for the value of λ_{BOR} that would exactly offset the common language boost between A and B.

12. Suppose over a decade countries A and C experience average annual real GDP growth rates of 2.5% and 4%, respectively, while D_{AC} is unchanged. Assume prices are stable and growth maps one-for-one into Y.

(a) Compute Y_A and Y_C after 10 years.

(b) Using the gravity model with $\beta_1 = \beta_2 = 1$, $\gamma = 1$, compute the growth factor for T_{AC} over the decade.

(c) If observed T_{AC} grows by only 12%, list two trade-cost or policy changes consistent with the shortfall relative to the gravity prediction.

13. Two countries, Home (H) and Foreign (F), produce Wine (W) and Cloth (C) with one input, labor. Unit labor requirements are:

	a_{LW}	a_{LC}
H	2	4
F	6	3

Labor endowments: $L_H = 240,\ L_F = 240.$

[1] Here, "border friction" means i and j are in different economic unions with additional administrative costs. Choose the signs so that a border *reduces* trade.

(a) Compute each country's autarky opportunity cost of W in terms of C and vice versa. Identify comparative advantage.

(b) Draw (or describe algebraically) each country's PPF. What is each country's autarky relative price P_W/P_C under competitive pricing equal to unit labor cost?

(c) Find the range of world relative prices P_W/P_C consistent with complete specialization.

(d) Suppose the world relative price is $P_W/P_C = 1.2$. Determine specialization, production bundles, and the pattern of trade.

(e) Compute the real wage in each country under free trade in terms of each good (units of W and units of C purchasable per unit of labor income).

14. Continue from Q1 with P_W/P_C moving from 1.2 to 0.8.

    (a) Re-evaluate which country (if any) changes specialization as the relative price falls to 0.8. Explain.

    (b) Compute the real wage of each country in terms of W and C at $P_W/P_C = 0.8$ and compare to autarky.

    (c) Which country's real wage benefits from an improvement in its terms of trade? Explain using the Ricardian logic.

15. There are three countries with the following unit labor requirements:

	a_{LW}	a_{LC}
1	1	2
2	2	3
3	4	5

$L_1 = L_2 = L_3 = 100.$

    (a) Order countries by comparative advantage in W relative to C.

    (b) Derive the world relative supply (WRS) schedule $RS^{W/C}(P_W/P_C)$ by progressively "turning on" countries into W as P_W/P_C rises. State the price thresholds where kinks occur.

    (c) If world relative demand is $RD^{W/C} = 1.1 - 0.2(P_W/P_C)$ (for $P_W/P_C \in [0, 5]$), solve for the free-trade equilibrium P_W/P_C and identify which countries produce which good.

16. In the two-country model of Q1, suppose Home experiences a neutral productivity improvement in Cloth only: a_{LC} falls from 4 to 3 (all else unchanged).

    (a) Re-compute Home's opportunity cost of W in terms of C and the specialization range.

    (b) Discuss how the shock shifts the world relative supply and the likely effect on the world relative price P_W/P_C.

    (c) With P_W/P_C adjusting to the new equilibrium, compare Home and Foreign real wages (in both goods) to the pre-shock free-trade case. Who gains more from this productivity growth?